

α -labeling of banana tree graphs

Borna Gojšić

Faculty of Electrical Engineering and Computing, Zagreb, Croatia

borna.gojsic@fer.hr

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A banana tree graph, denoted by $B(n, k)$, is a tree constructed by connecting n star graphs S_k to a root vertex. TODO: Add graceful labeling definition. An α labeling is a type of graceful labeling introduced by Rosa [1] which satisfies the α -inequality; there must exist a constant α such that for every edge uv , either one of the following holds:

- $f(u) \leq \alpha < f(v)$, or
- $f(v) \leq \alpha < f(u)$.

This article builds upon previous research on graceful labelings of banana trees and explores their α -labelability. We utilize the symmetries of the $B(n, k)$ trees to give the number of all of the non-isomorphic α -labelings based on specific subsets of them which we call *proper* and *central* α -labelings. We classify all the α -labelings for all the $B(n, 1)$, $B(1, k)$ and $B(2, k)$ trees. We present α -labelings of the $B(n, k)$ tree, for $k \geq \frac{n}{2} + 1$ and $n \in \mathbb{N}$. Finally, we prove that the $B(n, k)$ tree is not α -labelable for $1 < k \leq \frac{n}{2}$ and any $n \geq 3$, classifying all the $B(n, k)$ trees based on their α -labelability.

References

- [1] A. Rosa, On certain valuations of the vertices of a graph, in: Theory of graphs (Internat. Sympos., Rome, 1966), Gordon & Breach, New York, 1967, 349–355.